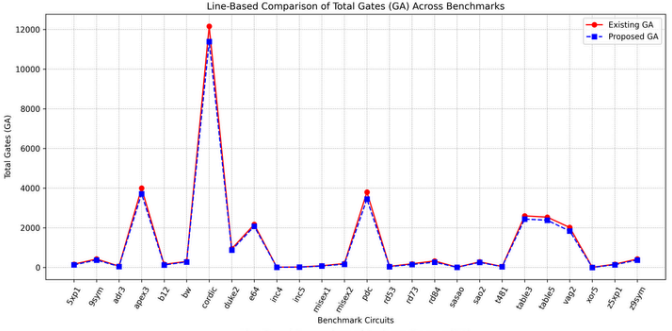
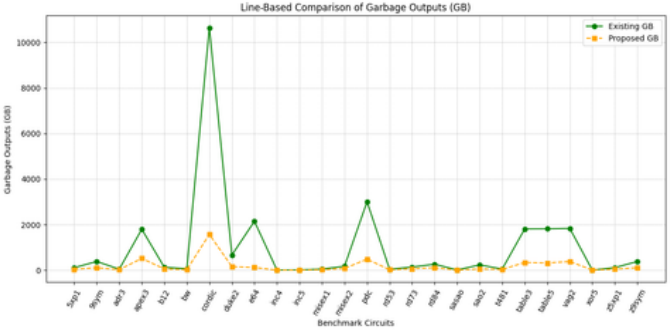
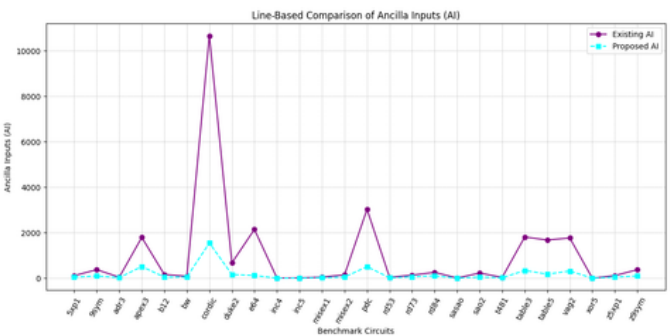

Line-Based Comparison of Total Gates (GA) Across Benchmarks
Existing GA
Proposed GA
Total Gates (GA)
0
2000
4000
6000
8000
10000
12000
5xp1
9sym
add3
apex3
b12
bw
cordic
duke2
ex4
inc4
inc5
misex1
misex2
pdc
rd53
rd73
rd84
sao4
sao2
t481
table3
table5
vg2
xor5
z5xp1
z9sym
Benchmark Circuits


Line-Based Comparison of Garbage Outputs (GB)
Existing GB
Proposed GB
Garbage Outputs (GB)
0
2000
4000
6000
8000
10000
5xp1
9sym
add3
apex3
b12
bw
cordic
duke2
ex4
inc4
inc5
misex1
misex2
pdc
rd53
rd73
rd84
sao4
sao2
t481
table3
table5
vg2
xor5
z5xp1
z9sym
Benchmark Circuits


Line-Based Comparison of Ancilla Inputs (AI)
Existing AI
Proposed AI
Ancilla Inputs (AI)
0
2000
4000
6000
8000
10000
5xp1
9sym
add3
apex3
b12
bw
cordic
duke2
ex4
inc4
inc5
misex1
misex2
pdc
rd53
rd73
rd84
sao4
sao2
t481
table3
table5
vg2
xor5
z5xp1
z9sym
Benchmark Circuits